Combinational Assignment

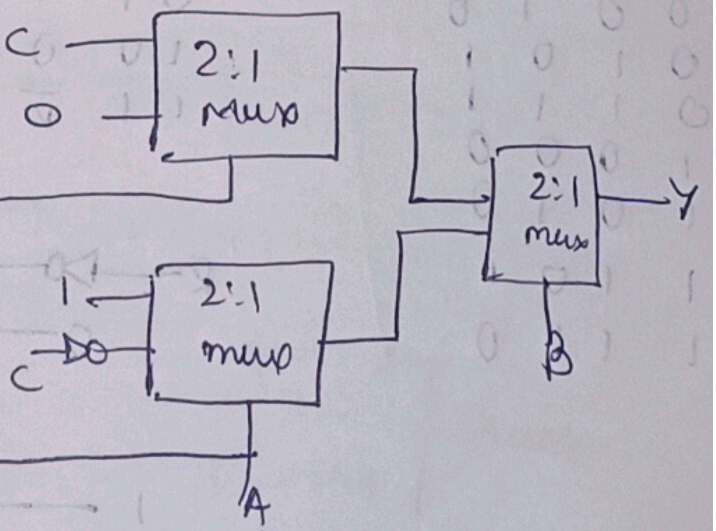
1) Implement the following functions using
i) 2:1 Mux and not gates
ii) 4:1 mux

a) $F(A,B,C) = \bar{A}C + B\bar{C}$
$= \bar{A}C(B+\bar{B}) + (A+\bar{A})B\bar{C}$
$= \bar{A}BC + \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$
$= m(1,2,3,6)$

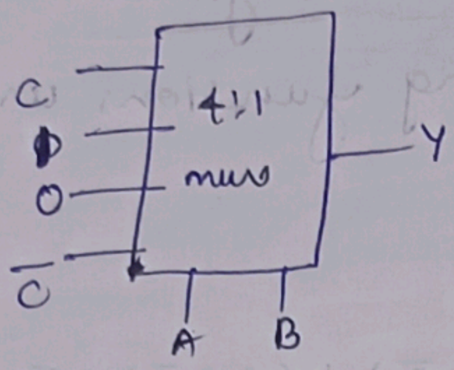
i) (using 2:1 mux and not gates)

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

A	B	Y
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$



ii) using 4:1 mux (follow the above truth table)



b) $F(P,q,R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

$= \bar{P}\bar{R}(q+\bar{q}) + (P+\bar{P})Q\bar{R} + \bar{P}Q(R+\bar{R})$

$= \bar{P}q\bar{R} + \bar{P}\bar{q}\bar{R} + PQ\bar{R} + \bar{P}Q\bar{R} + \bar{P}QR + \bar{P}Q\bar{R}$

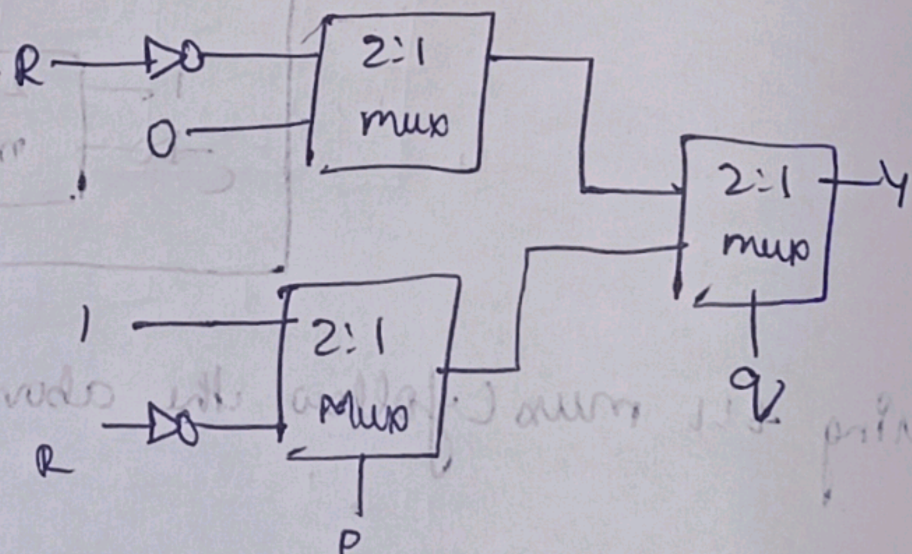
$= \bar{P}q\bar{R} + \bar{P}\bar{q}\bar{R} + pq\bar{R} + \bar{P}qR$

$Y = m(0,2,3,6)$

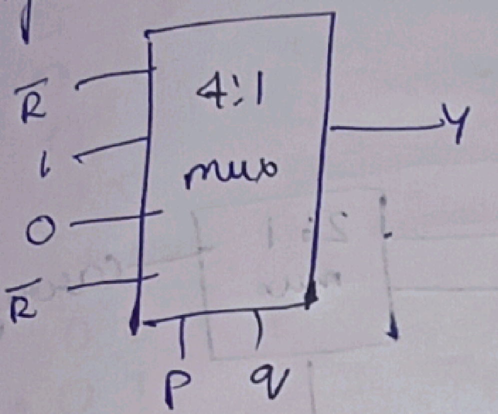
TT (using 2:1 mux and not gate)

P	q	R	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

P	q	Y
0	0	$\bar{R}$
0	1	1
1	0	0
1	1	$\bar{R}$



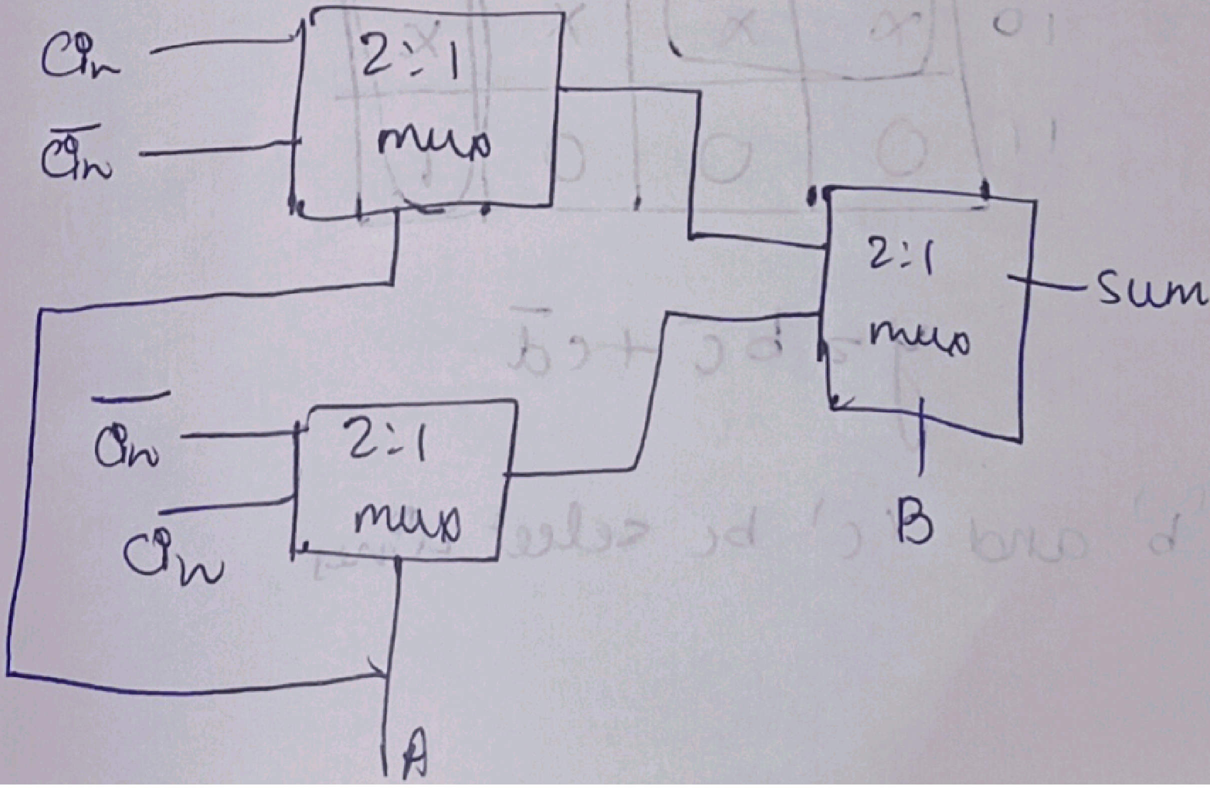
Q1) using 4:1 Mux

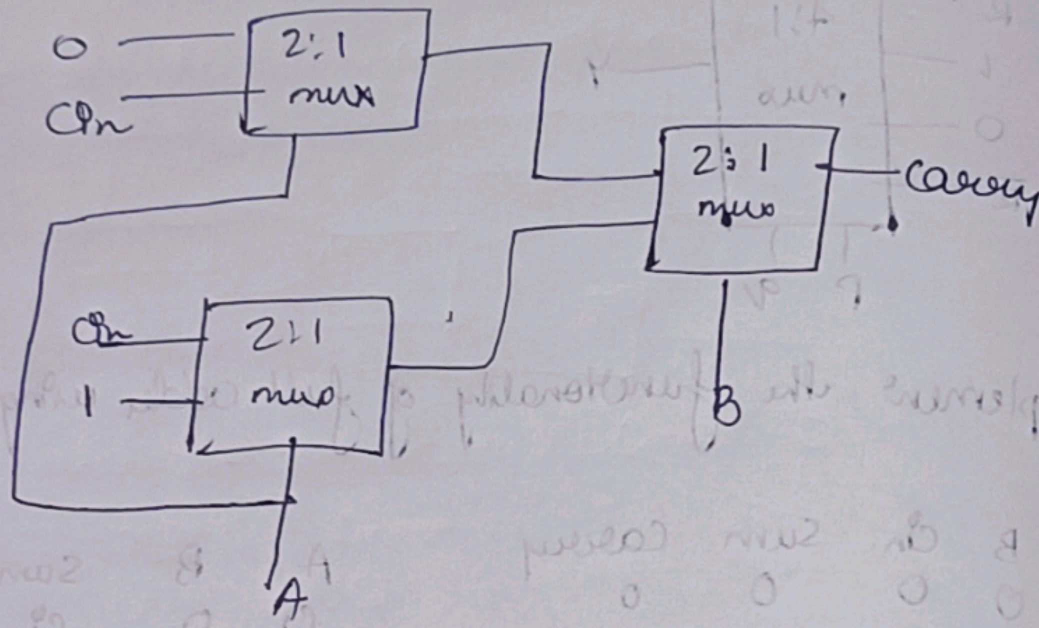


2) Implement the functionality of full adder using 2:1 Muxes only

A	B	C_{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

A	B	Sum	Carry
0	0	C_{in}	0
0	1	$\overline{C_{in}}$	C_{in}
1	0	$\overline{C_{in}}$	C_{in}
1	1	C_{in}	1





3) Simplify $f(a,b,c,d) = \Sigma m\,(2,4,5,6,10) + D(12, 13, 14, 15)$ and implement the SOP expression using 4:1 Mux

ab \ cd	00	01	10	11
00	0	0	0	1
01	1	1	0	1
10	X	X	X	X
11	0	0	0	1

$$y = b\bar{c} + c\bar{d}$$

Let 'b' and 'c' be select lines

b	c	d	$b\bar{c}$	$c\bar{d}$	$b\bar{c}+c\bar{d}$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

b	c	$b\bar{c}+c\bar{d}$
0	0	0
0	1	$\bar{d}$
1	0	1
1	1	$\bar{d}$